Monolingual Results

Language	Pair n-Gram Model (baseline 2020)	Encoder-decoder Transformer (baseline 2020)	Neural Transducer (baseline 2021/22)	Attentive LSTM (baseline 2024)
Adyghe	27.00	39.00	20.00	30.00
Arabic	43.00	43.00	53.00	45.00
Armenian (Eastern)	17.00	16.00	15.00	20.00
Assamese	14.00	15.00	7.00	12.00
Belarusian	7.00	6.00	2.00	2.00
Bengali	70.00	78.00	68.00	67.00
Bulgarian	37.00	30.00	32.00	27.00
Burmese	39.00	41.00	29.00	35.00
Cebuano	22.00	26.00	20.00	20.00
Central Khmer	56.00	43.00	31.00	35.00
Dutch	32.00	29.00	24.00	23.00
Eastern Lawa	44.00	24.00	8.00	12.00
English	70.00	70.00	63.00	66.00
French	31.00	29.00	23.00	27.00
Georgian	0.00	4.00	0.00	1.00
German	49.00	56.00	46.00	45.00
Greek	27.00	26.00	20.00	27.00
Hindi	23.00	24.00	11.00	13.00
Hungarian	9.00	11.00	7.00	8.00
Icelandic	35.00	24.00	12.00	16.00
Indonesian	58.00	52.00	64.00	73.00
Irish	57.00	46.00	43.00	39.00
Italian	22.00	21.00	15.00	16.00
Japanese (Hiragana)	23.00	20.00	10.00	12.00
Korean	81.00	89.00	23.00	100.00
Latvian	51.00	51.00	51.00	50.00
Lithuanian	32.00	35.00	33.00	31.00
Macedonian	6.00	5.00	5.00	5.00
Maltese (Latin)	27.00	24.00	17.00	22.00
Norwegian Nynorsk	61.00	69.00	66.00	69.00
Pashto	70.00	68.00	67.00	66.00
Persian (Classical)	51.00	58.00	57.00	51.00
Persian (Iranian)	66.00	63.00	65.00	65.00
Polish	9.00	8.00	4.00	7.00
Romanian	10.00	17.00	9.00	17.00
Russian	31.00	21.00	23.00	25.00
Serbo-Croatian (Latin)	84.00	69.00	64.00	69.00
Shan	6.00	10.00	5.00	4.00
Slovenian	73.00	52.00	56.00	50.00
Spanish	3.00	10.00	4.00	3.00
Swedish	67.00	68.00	59.00	61.00
Tagalog	11.00	17.00	13.00	11.00

Language	Pair n-Gram Model (baseline 2020)	Encoder-decoder Transformer (baseline 2020)	Neural Transducer (baseline 2021/22)	Attentive LSTM (baseline 2024)
Thai	70.00	49.00	39.00	42.00
Ukrainian	27.00	26.00	19.00	18.00
Urdu	67.00	66.00	72.00	77.00
Uyghur	0.00	3.00	0.00	2.00
Vietnamese (Hanoi)	44.00	20.00	5.00	4.00
Welsh	33.00	20.00	12.00	19.00
Welsh (Southern Dialect)	28.00	18.00	13.00	13.00
Macro-average WER	**37.14**	**34.88**	**28.65**	**31.67**

Error Analysis

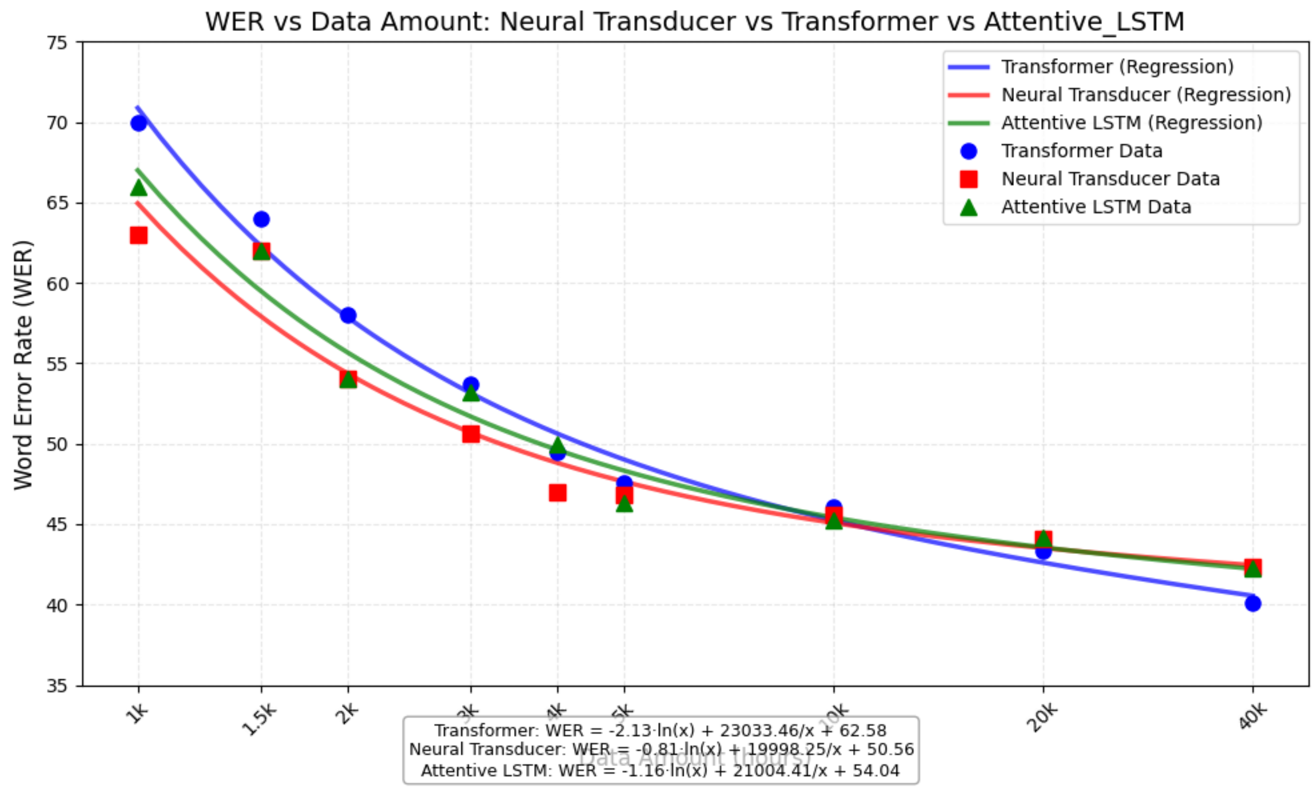
Korean

	Neutral Transducer	Encoder-Decoder Transformer	Attentive LSTM
Hangul	23	89	100
Jamo	23	28	28

How Data Amount Affects Performance of Transformer Model

Comparing best performing model and transformer on different amount fo English data

Data Amount	Neutral Transducer	Encoder-Decoder Transformer	Attentive LSTM
1k	63.00	70.00	66.00
1.5k	62.00	64.00	62.00
2k	54.00	58.00	54.00
3k	50.67	53.67	53.20
4k	47.00	49.50	50.00
5k	46.80	47.60	46.30
10k	45.60	46.10	45.21
20k	44.05	43.35	44.13
40k	42.38	40.10	42.23



Multilingual Approach

A tag comprising a language code (e.g. <|ENG|>, <|GER|>) is prepended to each grapheme sequence source. The datasets of each language are simply concatenated.

Does converting the language tag to the corresponding script of the language improve the model performance?

Using `slavic_cyrillic` dataset we got following validation WER:

Tag in	Neural Transducer	Encode-Decoder Transformer	Attentive LSTM
Uppercase Cyrillic	39.67	28.00	28.00
Uppercase Latin	31.00	28.00	27.00
Lowercase Latin	32.33	28.00	27.00
No tag	45.00	44.67	44.67

Results

Same Language Family and Same Script

			Transducer			**Transformer**			**Attentive LSTM**		
Family	Script	Language	Multi	Mono	Δ	Multi	Mono	Δ	Multi	Mono	Δ
Germanic	Latin	German	48	46	2	47	56	-9	42	45	-3
		Dutch	25	24	1	21	29	-8	20	23	-3
		Swedish	62	59	3	60	68	-8	60	61	-1
Average					2.00			-8.33			-2.33
Romance	Latin	Italian	26	15	11	22	21	1	18	16	2
		Spanish	12	4	8	3	10	-7	2	3	-1
		Romanian	18	9	9	13	17	-4	13	17	-4
Average					9.33			-3.33			-1.00
Slavic	Cyrillic	Russian	27	23	4	18	21	-3	22	25	-3
		Ukrainian	22	19	3	27	26	1	19	18	1
		Bulgarian	35	32	3	27	30	-3	31	27	4
Average					3.33			-1.67			0.67

			Transducer			Transformer			Attentive LSTM		
		Serbo-Croatian (Latin)	67	64	3	63	69	-6	68	69	-1
Slavic	Latin	Polish	13	4	9	9	8	1	9	7	2
		Slovenian	59	56	3	52	52	0	50	50	0
Average					5.00			-1.67			0.33
		Pashto	79	67	12	66	68	-2	64	66	-2
Indo-Iranian	Arabic	Persian (Iranian)	73	65	8	67	63	4	64	65	-1
		Urdu	78	72	6	64	66	-2	71	77	-6
Average					8.67			0.00			-3.00
		Indonesian	66	64	2	57	52	5	55	73	-18
Austronesian	Latin	Cebuano	26	13	13	24	26	-2	22	20	2
		Tagalog	15	20	-5	18	17	1	16	11	5
Average					3.33			1.33			-3.67

Different Language Family and Same Script

			Transducer			Transformer			Attentive LSTM		
Family	Script	Language	Multi	Mono	Δ	Multi	Mono	Δ	Multi	Mono	Δ
Afro-Asiatic		Arabic	57	53	4	38	43	-5	53	45	8
Indo-European	Arabic	Persian (Classic)	62	57	5	45	58	-13	55	51	4
Turkic		Uyghur	8	0	8	2	3	-1	4	2	2
Average					5.67			-6.33			4.67
Indo-European		Italian	19	15	4	20	21	-1	17	16	1
Afro-Asiatic	Latin	Maltese (Latin)	18	17	1	21	24	-3	15	22	-7
Uralic		Hungarian	6	7	-1	8	11	-3	3	8	-5
Average					1.33			-2.33			-3.67

Larger Model/Dataset

			Neural Transducer			Encoder-decoder Transformer			Attentive LSTM		
Family	Script	Language	Multi	Mono	Δ	Multi	Mono	Δ	Multi	Mono	Δ
Indo-European		Italian	16	15	1	22	21	1	19	16	3
Afro-Asiatic		Maltese (Latin)	19	17	2	18	24	-6	17	22	-5
Uralic	Latin	Hungarian	9	7	2	12	11	1	7	8	-1
Austronesian		Indonesian	56	64	-8	52	52	0	57	73	-16
Average					-0.75			-1.00			-4.75

Same Language Family and Different Script

			Transducer			Transformer			Attentive LSTM		
Family	Script	Language	Multi	Mono	Δ	Multi	Mono	Δ	Multi	Mono	Δ
	Armenian	Armenian (Eastern)	20	15	5	14	16	-2	18	20	-2
Indo-European	Greek	Greek	21	20	1	25	26	-1	23	27	-4
	Cyrillic	Russian	22	23	-1	21	21	0	20	25	-5
Average					1.67			-1.00			-3.67

Larger Model/Dataset

			Neural Transducer			Encoder-decoder Transformer			Attentive LSTM		
Family	Script	Language	Multi	Mono	Δ	Multi	Mono	Δ	Multi	Mono	Δ
Indo-European	Armenian	Armenian (Eastern)	15	15	0	17	16	1	19	20	-1
	Greek	Greek	22	20	2	26	26	0	25	27	-2

			Neural Transducer			Encoder-decoder Transformer			Attentive LSTM		
	Cyrillic	Russian	22	23	-1	26	21	5	24	25	-1
	Latin	Italian	19	15	4	21	21	0	21	16	5
Average					1.25			1.50			0.25

Effect of Transliteration

Latin transliteration of languages with Cyrillic script

Monolingual

	Neural Transducer			Transformer			Attentive LSTM		
	Latin	Cyrillic	Δ	Latin	Cyrillic	Δ	Cyrillic	Latin	Δ
Russian	20	23	-3	22	21	1	30	25	5
Ukrainian	22	19	3	24	26	-2	20	18	2
Bulgarian	34	32	2	35	30	5	29	27	2
Macedonian	6	5	1	5	5	0	5	5	0
Average			0.75			1.00			2.25

Multilingual (Slavic Latin) Dataset

			Neural Transducer			Encoder-decoder Transformer			Attentive LSTM		
Family	Script	Language	Multi	Mono	Δ	Multi	Mono	Δ	Multi	Mono	Δ
		Bulgarian*	31	34	-3	27	35	-8	23	27	-4
		Russian*	19	20	-1	24	22	2	20	25	-5
		Ukrainian*	23	22	1	18	24	-6	19	18	1
Slavic	Latin	Serbo-Croatian (Latin)	76	64	12	63	69	-6	61	69	-8
		Polish	13	4	9	7	8	-1	8	7	1
		Slovenian	54	56	-2	57	52	5	49	50	-1
		Macedonian*	4	6	-2	3	5	-2	4	5	-1
Average					2.00			-2.29			-2.43

Reference Result from previous shared task

2022

Model

Baseline: A neural transducer system using an imitation learning paradigm (dyNET framework)

Submissions:

1. Tü-G2P: A series of sequence labelling systems to G2P tasks, which use simpler alignment rather than dynamic transducer-based alignment.(Pytorch)
2. Hammond (Repo): A non-neural system based on OpenFST and uses weighted finite-state transducers and expectation-maximization to compute the best many-to-many alignment of letters and phonetic symbol
3. ~~mSLAM: Non-archival; abstract only;~~ useless
4. ~~NFST: Non-archival; abstract only;~~ useless

Reference Result:

- Baseline

Language	Bengali	Burmese	German	Irish	Italian	Persian	Swedish	Tagalog	Thai	Ukrainian	Macro-average
WER	67.12	29.00	42.00	38.00	15.00	59.65	45.00	20.00	21.00	32.00	36.88

- Hammond (trigram alignment)

Language	Bengali	Burmese	German	Irish	Italian	Persian	Swedish	Tagalog	Thai	Ukrainian	Macro-average
WER	68.49	48.00	61.00	51.00	25.00	67.86	55.00	18.00	72.00	50.00	51.63

2024

Model

Baseline: attentive_gru, attentive_lstm, gru, hard_attention_gru, lstm, pointer_generator_gru, transducer_gru, transformer(20\40\60 epochs)

Reference Result:

- Baseline

Models	gru	lstm	attentive_gru	attentive_lstm	hard_attention_gru	hard_attention_gru (Arab)	pointer_generator_gru	transducer_gru	transforme
WER (%)	43.75	44.25	63.08	47.25	40.67	31.33	62.17	69.33	78.25

- Best performance model on all datasets(hard_attenton_gru)

Languages	Arabic	Bulgarian	English	Persian	Indonesian	Macedonian	Pashto	Russian	Spanish	Tagalog	Ukrainian	Urdu
WER	31.33	20.00	58.00	29.67	55.33	3.67	44.33	10.33	5.00	40.33	15.67	64.00

- GRU、LSTM、Transformer on different languages

Model/Languages	English	Pashto	Russian	Spanish
GRU	31.00	39.00	14.00	9.00
LSTM	48.33	57.67	10.33	9.00
Transformer	81.33	77.00	35.67	24.67

- Comparison between hard_attention_gru、attentive_lstm

Model / Language	Korean	Bengali	Indonesian	Pashto	Swedish
hard_attention_gru	98.00	87.00	85.00	78.00	66.00
attentive_lstm	100.00	67.00	73.00	66.00	61.00

- Comparison between attentive_gru and attentive_lstm

The comparison of validation accuracy (val_accuracy) between attentive_gru (GRU) and attentive_lstm (LSTM) across all languages shows:

- LSTM val_accuracy higher than GRU: 20 times
- LSTM val_accuracy lower than GRU: 22 times
- LSTM val_accuracy equal to GRU: 6 times
- Average LSTM val_accuracy: 0.6987
- Average GRU val_accuracy: 0.7049

This indicates that, on average, attentive_gru achieves slightly higher validation accuracy than attentive_lstm, although the performance varies across different languages.

- attentive_lstm performance on Adyghe and Bengali with different parameters(DEV_WER)

Params	Adyghe WER	Bengali WER	Dutch WER	Urdu WER
default	31.00	65.00	23.00	77.00
4 encoder_layers + 1 decoder_layer	51.00	70.00	45.00	80.00
256 embedding_size + 1024 hidden_size	37.00	67.00	30.00	78.00
0.1 label_smoothing	33.00	65.00	21.00	75.00